FPS TUNE-UP SERIES

RTX 40 & 50 SERIES LAPTOP EDITION

Gaming PC Optimization Package

A plain-English, step-by-step tune-up for new gaming laptop owners. Smoother gameplay, fewer stutters, and the right settings, without the jargon.

- Built from a real Legion + RTX 5070 tune-up
- NVIDIA settings explained
- Skyrim SE / Nolvus
- Fallout 4
- Cyberpunk 2077
- 15-minute Quick-Start Card
- Before/after checklist
- Glossary

$99
COMPLETE GUIDE

Reference system: Lenovo Legion · NVIDIA GeForce RTX 5070 Laptop GPU (8 GB) · Intel Core Ultra 9 · 32 GB RAM · 1080p display (1440p external monitor notes included). Works for any RTX 40 or 50 Series gaming laptop.

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0 Start Here

How to use this guide

This guide is for people who just bought a gaming laptop and want it to run games the way it should, **without** needing to be a PC expert. Every step is numbered, every setting tells you **exactly what to pick**, and every technical word is explained the first time it appears (and again in the **Glossary** at the back).

It was built from a real tune-up of a **Lenovo Legion** laptop with an **NVIDIA GeForce RTX 5070 Laptop GPU** (8 GB of video memory), an **Intel Core Ultra 9** processor, and **32 GB of RAM**, playing on its built-in 1080p screen. Lenovo-specific steps are clearly marked. If you own a different brand with an RTX 40 or 50 Series graphics card, every NVIDIA and Windows step still applies. Just use your brand's own control app wherever we mention Lenovo Vantage or Legion Space.

Four words come up constantly: GPU *(the graphics card that draws the game)*, CPU *(the processor, the laptop's main "brain")*, FPS *(frames per second: higher = smoother)*, and VRAM *(the graphics card's own memory, 8 GB on an RTX 5070 Laptop GPU)*.

How long it takes

- **Full guide:** about 60–90 minutes the first time (mostly waiting on downloads).
- **Quick-Start Card:** about 15 minutes once updates are done.

What you need

- Laptop **plugged into its charger**
- An internet connection
- An administrator (owner) Windows account
- About 10–20 GB of free disk space for updates

How each chapter is laid out

Tip: a helpful shortcut **In plain English**: why it matters **Note**: extra info **Caution**: slow down

Stop: do not skip

Each chapter header also shows a **time estimate** and a **difficulty** level. Tip: fill in the **"Before"** column of the Chapter 8 checklist **first**, so you can see what improved.

SAFETY FIRST: MAKE A RESTORE POINT

A restore point *(a saved snapshot of Windows settings you can roll back to)* lets you undo changes if something goes wrong.

1. Press the **Windows key**, type **Create a restore point**, and press Enter.
2. Select your **C:** drive. If protection is "Off", click **Configure** → **Turn on system protection** → **OK**.
3. Click **Create...**, name it "Before tune-up", and click **Create**.

1 Driver and BIOS Updates

Get the foundation right before touching any settings

Difficulty: Easy | Time: 20–40 min | Restart required

IN PLAIN ENGLISH

New laptops often ship with drivers that are months old. NVIDIA regularly releases Game Ready drivers *(graphics drivers tuned for newly released games)* that fix bugs and stutters. The BIOS *(the built-in firmware that starts your laptop before Windows loads)* controls things like fan behavior and power limits, so its updates can matter too.

1.1 Update the NVIDIA graphics driver

Option A: the easy way (recommended for most people)

1. Download and install the **NVIDIA app** from nvidia.com (it replaces the older GeForce Experience).
2. Open it and go to **Drivers**. Make sure **Game Ready Driver** is selected (not Studio Driver).
3. Click **Download**, then **Express Installation**. The screen may flicker or go black for a few seconds. That's normal.
4. Restart the laptop when it finishes.

Option B: a clean install (use this if games crash, the driver seems broken, or you see unexplained GPU usage like the issue in Chapter 2.6)

1. On nvidia.com, go to **Drivers** and pick **GeForce** → **RTX 50 Series (Notebooks)** or **RTX 40 Series (Notebooks)** → your exact GPU → **Windows 11** → **Game Ready**. Download it.
2. Run the installer, choose **Custom (Advanced)**, then tick **Perform a clean installation**.
3. Finish the install and restart.

TIP

A clean installation resets your NVIDIA settings to default. That's fine: Chapter 2 tells you what to set afterward.

1.2 Update the BIOS and chipset (Lenovo Vantage or your brand's app)

The chipset driver *(software that helps Windows manage the motherboard, power, and ports)* and the BIOS come from the **laptop maker**, not NVIDIA.

1. **Plug in the charger.** Never do this step on battery.
2. Open **Lenovo Vantage** (or **Legion Space** on newer Legion models). Other brands: use the brand's control app or the support page for your model.
3. Go to **System Update** → **Check for updates**.
4. Install **Critical** and **Recommended** updates, including BIOS, chipset, and Intel drivers if listed.
5. Let the laptop restart on its own. A BIOS update can show a progress bar and restart more than once.

⚠ **NEVER INTERRUPT A BIOS UPDATE**

- Keep the laptop **plugged in** the whole time.
- **Do not** close the lid, press the power button, or unplug it, even if the screen looks frozen for several minutes.
- **Back up** important files first (cloud or external drive). A failed BIOS update is rare, but it can stop the laptop from starting.

1.3 Run Windows Update

1. Open **Settings** (`Windows key + I`) → **Windows Update** → **Check for updates**.
2. Install everything offered and restart.
3. Under **Advanced options** → **Optional updates**, skip any **display/NVIDIA** driver that's older than the one you installed.

⚠ **ORDER MATTERS**

Do Windows Update and BIOS/chipset **first** and the NVIDIA driver **last**. That way Windows can't replace your fresh NVIDIA driver with an older one.

2 NVIDIA Control Panel & NVIDIA App

The settings that make the biggest difference to smoothness

Difficulty: Easy–Medium | Time: 10–15 min

NVIDIA gives you **two** programs. The **NVIDIA Control Panel** holds the classic 3D settings. The **NVIDIA app** holds newer features like DLSS overrides and the overlay. You'll use both.

2.1 Global 3D settings

1. Right-click the desktop → **Show more options** → **NVIDIA Control Panel** (or search for it in Start).
2. Click **Manage 3D settings** → **Global Settings** tab.
3. Change the settings in the table below, then click **Apply** at the bottom right.

Setting	Set it to	Why (plain English)
Power management mode	**Prefer maximum performance**	Stops the graphics card from dropping to low speed mid-game, which causes dips. Uses a little more power while gaming.
Shader Cache Size	**10 GB (Unlimited if you have lots of free disk space)**	A shader *(a small program that tells the GPU how to draw lighting and surfaces)* must be prepared before use. The shader cache *(a folder of already-prepared shaders)* saves that work so games stutter less next time.
Texture filtering – Quality	**High performance**	Slightly faster texture processing. The visual difference is very small.
Texture filtering – Negative LOD bias	**Allow**	LOD bias *(how sharp distant textures look)*. "Allow" lets games and DLSS keep textures sharp as intended.
Texture filtering – Trilinear optimization	**On**	A small, safe speed-up in how textures blend.
Low Latency Mode	**On**	Latency *(the delay between pressing a button and seeing it happen)*. "On" trims that delay. Use **Ultra** only for games with **no** NVIDIA Reflex option (see 2.2).
Monitor Technology	**G-SYNC / G-SYNC Compatible**	Turns on smooth variable refresh (explained in 2.3).
Vertical sync	**On**	Works with G-Sync as a safety net against screen tearing (see 2.3).
Preferred refresh rate	**Highest available**	Makes sure games use your screen's fastest refresh rate.
Threaded optimization	**Auto**	Lets the driver decide when to spread work across CPU cores. Auto is best for almost every game.

2.2 Reflex vs. Low Latency Mode

NVIDIA Reflex *(a lag-reducing feature built directly into many modern games)* works better than the driver's Low Latency Mode because the game itself controls it.

- **Game has Reflex?** Turn **Reflex On** in the game's settings. Leave the driver's Low Latency Mode at **On**.
- **Game has no Reflex?** Set Low Latency Mode to **Ultra** for that game only (per-game profiles are in Chapter 3).

2.3 The smooth-screen recipe: G-Sync + V-Sync + frame cap

Your screen's refresh rate *(how many times per second the screen redraws, measured in Hz)* is probably 144–240 Hz. G-Sync *(NVIDIA's version of VRR, variable refresh rate: the screen waits for each new frame instead of redrawing on a fixed timer)* removes tearing and judder. V-Sync *(vertical sync: holds frames until the screen is ready, preventing "tearing", a horizontal split in the image)* backs it up.

#	Where	What to set
1	NVIDIA Control Panel → **Set up G-SYNC**	Tick **Enable G-SYNC / G-SYNC Compatible** → **Enable for full screen mode** (or full screen and windowed). Apply.
2	Manage 3D settings → Vertical sync	**On**
3	Each game's video settings	V-Sync **Off**
4	Frame cap: Reflex, the game's own limiter, or **Max Frame Rate** in Manage 3D settings	About **3 fps below** your refresh rate: 162 on a 165 Hz screen, 141 on 144 Hz, 237 on 240 Hz. Reflex + G-Sync + V-Sync caps automatically, so a manual cap is mainly for games without Reflex.

DON'T SEE "SET UP G-SYNC"?

On many laptops the built-in screen only supports G-Sync in **dGPU-only / Discrete** mode (Chapter 4.4) or on models with **Advanced Optimus**. Switch modes, restart, and check again.

2.4 DLSS overrides in the NVIDIA app

DLSS *(Deep Learning Super Sampling: the game draws at a lower resolution and the RTX card's AI rebuilds a sharp full-resolution image, giving more FPS)* has several parts. The NVIDIA app can force the newest version even in games that shipped with an older one.

1. Open the **NVIDIA app** → **Graphics** → **Global Settings** (or choose one game under **Program Settings**).
2. Find the **DLSS Override** options and set them as below.

DLSS override	Set it to	Plain-English meaning
Model presets / DLSS version	**Latest**	Always use NVIDIA's newest DLSS model.
Super Resolution preset	**Preset K**	Super Resolution *(the upscaling part of DLSS)*. Preset K is a newer, sharper, more stable model.
Frame Generation	**2x**	Frame Generation *(AI creates an extra frame between each pair of real frames)*. Only use it when the game already runs at **about 60 fps or more without it**; below that it can feel laggy.

DLSS override	Set it to	Plain-English meaning
Ray Reconstruction	**On / Latest where supported**	Ray Reconstruction *(AI that cleans up noisy ray-traced lighting)*. Only works with ray tracing *(realistic simulated light and reflections)* turned on.

RTX 50 Series laptops can also do 3x/4x "Multi Frame Generation". Start with **2x**: the least lag and fewest glitches. Frame Generation also needs **HAGS** on (Chapter 4.3).

2.5 Turn off features you don't need

Feature (NVIDIA app)	Set to	Why
System → Performance → Automatic Tuning	**Off**	This is an automatic overclock. Leave it off for stability.
System → Video → RTX Video Super Resolution	**Off if unused**	Upscales web videos using the GPU. Nice for streaming, but it uses GPU power.
System → Video → RTX Video HDR / RTX HDR	**Off if unused**	Converts video or games to HDR. Not needed on most laptop screens.
Settings → Features → NVIDIA Overlay	**Off if unused**	The in-game overlay (Alt+Z). Turn it on temporarily when measuring FPS (Chapter 8).
Overlay → Instant Replay	**Off if unused**	Constantly records the last few minutes of gameplay in the background.

2.6 Fix: the GPU is busy when nothing is running

> ⚠ **REAL FINDING FROM THE REFERENCE TUNE-UP**
> On the test laptop, an NVIDIA background process named `nvgwls` was using **25–34% of the GPU while the laptop sat idle** with no game open. That's power, heat, and fan noise wasted before you even start playing.

How to check yours: press `Ctrl + Shift + Esc` to open **Task Manager** → **Processes**. Right-click the column headers and tick **GPU** if it isn't shown, then click **GPU** to sort. With no games open, nothing should be using a large share.

How to fix it (try in order):

1. Turn off **NVIDIA Overlay**, **Instant Replay**, the **RTX Video** features, and **Automatic Tuning** (table 2.5).
2. **Restart** the laptop and check Task Manager again.
3. Still high? Do a **clean driver install** (Chapter 1.1, Option B), then re-apply this chapter's settings.

2.7 Using a 1440p external monitor

- Plug into the **HDMI** or **USB-C / Thunderbolt** port wired to the NVIDIA card (your manual lists which; on Legion models these are usually the rear ports).
- In **Settings → System → Display → Advanced display**, select the monitor and choose its **highest refresh rate**.
- 1440p *(2560×1440, about 78% more pixels than 1080p)* is harder to run: use DLSS **Balanced** or **Performance**, and drop **texture quality** one step if 8 GB of VRAM causes stutters.

3 Per-Game Profiles

Skyrim SE / Nolvus · Fallout 4 · Cyberpunk 2077

Difficulty: Medium | Time: 5–10 min per game

A per-game profile *(settings for one game only, overriding the global ones)* lets you fine-tune a single game without affecting the rest.

3.0 How to create a profile

1. NVIDIA Control Panel → **Manage 3D settings** → **Program Settings** tab.
2. Click **Add** and pick the game, or **Browse** to its file: `SkyrimSE.exe`, `Fallout4.exe`, or `Cyberpunk2077.exe`.
3. Change only the settings listed for that game below, then click **Apply**.

3.1 Skyrim Special Edition / Nolvus modlist

Nolvus is a huge, pre-built modlist *(a curated bundle of hundreds or thousands of mods that installs together)* for Skyrim SE. It is very **VRAM-heavy**: high-resolution textures are the main thing that overloads an 8 GB card.

Setting	Recommended	Why
Texture resolution (installer option)	**2K**	On 8 GB cards, 4K texture packs overflow VRAM and cause heavy stutter. 2K looks great at 1080p.
Power management mode (profile)	**Prefer maximum performance**	Prevents the GPU from slowing down in quieter scenes.
Frame cap	**The modlist's limiter**	Skyrim's physics can misbehave at very high FPS. Cap with the ENB *(a graphics add-on that changes lighting and effects)* frame limiter or the modlist's display settings, at the value the Nolvus guide recommends.
Everything else	**Modlist defaults**	Nolvus is tested as a package. Use its recommended in-game and ENB settings rather than random internet tweaks.

✓ **TIP**

Install Nolvus to a **short folder path** like `C:\Nolvus` (see Chapter 6). Long paths and folders inside Program Files or OneDrive cause install errors.

3.2 Fallout 4

⚠ **DON'T UNCAP FALLOUT 4**

Fallout 4's **physics and game speed are tied to frame rate**. Running too fast (above roughly 144 fps) can cause speed-up bugs, jittery objects, and broken scripts. Keep it limited.

1. Open `Documents\My Games\Fallout4\Fallout4Prefs.ini` in Notepad.

2. Under `[Display]`, make sure the line reads `iPresentInterval=1`. This is Fallout 4's built-in V-Sync; keep it on. Save the file.
3. In the NVIDIA profile for `Fallout4.exe`, set **Max Frame Rate** to **141 or lower**, so a 165 or 240 Hz screen can't push the game past ~144 fps.
4. If you see odd physics (shaking objects, lockpicking too fast), lower the cap further. Many players use 60.

> **NOTE**
> Fallout 4 has no NVIDIA Reflex, so you can set **Low Latency Mode: Ultra** in its profile. The game's launcher can rewrite the INI file, so re-check `iPresentInterval` after changing options there.

3.3 Cyberpunk 2077

In-game setting	Recommended	Why
DLSS Super Resolution	**Quality (1080p) · Balanced (1440p)**	A big FPS boost with little visual loss. Try Balanced at 1080p if you need more FPS.
DLSS Frame Generation	**On**	Only if you're already near 60 fps without it (see 2.4).
NVIDIA Reflex	**On**	Lowers input lag. Turned on automatically with Frame Generation.
Ray Tracing	**Optional; start Low/ Medium**	Looks great but costs a lot of FPS and VRAM. Raise it only if you have FPS to spare.
Ray Reconstruction	**On (with RT on)**	Cleaner, sharper ray-traced lighting.
V-Sync (in game)	**Off**	The NVIDIA Control Panel handles V-Sync (2.3).
Texture Quality	**Medium if stuttering**	Helps 8 GB cards, especially with ray tracing or at 1440p.

> **⚠ USING MODS THROUGH VORTEX?**
> Vortex *(a popular mod manager app)* makes modding easy, but some mods, especially after a game patch, can break DLSS, Frame Generation, or Reflex (options greyed out, crashes, or no FPS gain). After installing or updating mods, check those options still appear and work. If not, disable recent mods in Vortex one at a time to find the cause.

4 Windows & Laptop Power Settings

Make sure the laptop is allowed to run at full speed

Difficulty: Easy | Time: 10 min | One restart (GPU mode)

IN PLAIN ENGLISH

Gaming laptops hold back power by default to stay quiet and save battery. These settings tell Windows and the laptop: "When I'm plugged in and gaming, go full speed."

4.1 Windows power plan

Situation	Use	Where
Plugged in, gaming	**Ultimate Performance or High Performance (or Power mode: Best performance)**	Control Panel → Power Options, or Settings → System → Power & battery → Power mode
On battery	**Balanced + Battery saver**	Settings → System → Power & battery → Battery saver

To unlock "Ultimate Performance" (optional): right-click Start → **Terminal (Admin)**, paste the line below, press Enter, then pick the new plan in **Control Panel → Power Options**.

```
powercfg -duplicatescheme e9a42b02-d5df-448d-aa00-03f14749eb61
```

ONLY SEE "BALANCED"?

Many modern laptops hide the classic plans. That's OK: set **Power mode** to **Best performance** in Settings when plugged in. The Legion's own Performance mode (4.4) matters more anyway.

4.2 Game Mode

1. Settings → **Gaming** → **Game Mode** → **On**. It tells Windows to prioritize your game and hold off background jobs like update installs.

4.3 Hardware-accelerated GPU scheduling (HAGS)

HAGS *(lets the graphics card manage its own work queue instead of the CPU)* is **required** for DLSS Frame Generation.

1. Settings → System → **Display** → **Graphics** → **Advanced graphics settings** (called "Change default graphics settings" on some versions).
2. Turn **Hardware-accelerated GPU scheduling On**. Also turn on **Optimizations for windowed games** and **Variable refresh rate** if you see them.
3. Restart.

4.4 Lenovo Vantage / Legion Space settings LENOVO

Thermal mode: Performance

Press `Fn + Q` repeatedly until the **power button light turns red**. That's Performance mode. (White = Balanced, blue = Quiet; some models add purple = Custom.) You can also pick it in Vantage / Legion Space. Full Performance mode needs the charger plugged in.

GPU working mode

Mode	Best for	What it does
dGPU only / Discrete	**Maximum gaming performance**	The NVIDIA card drives the screen directly. Usually enables G-Sync on the built-in screen. Uses more battery.
Hybrid	**Battery life, school/work**	Uses Intel graphics for everyday tasks and the NVIDIA card for games. Slightly lower game performance.

Here dGPU *(discrete GPU: the separate NVIDIA card)* is compared with iGPU *(integrated GPU: small graphics built into the Intel processor)*. The switch is a MUX switch *(hardware that chooses which GPU drives the screen)*, so **changing GPU mode needs a restart**.

Overclocking: leave it off

Vantage / Legion Space may offer GPU or CPU overclocking *(running chips faster than stock speed)*. Beginners should **leave it off**: the gains are small, and it can cause crashes and extra heat.

> ✓ **TIP: ALWAYS GAME PLUGGED IN**
>
> On battery, a gaming laptop cuts GPU power sharply, and many lower the screen's refresh rate. Plug in for every gaming session.

5 Background App Cleanup

Free up the CPU, GPU, and memory for your game

Difficulty: Easy | Time: 5–10 min

IN PLAIN ENGLISH

Every app running in the background takes a small slice of your laptop's power. Ten small slices add up to stutters and lower FPS. This is one of the most common reasons a new laptop under-performs.

5.1 Stop unneeded apps from starting with Windows

1. Press `Ctrl + Shift + Esc` to open **Task Manager** → **Startup apps**.
2. Sort by **Startup impact**. Right-click anything you don't need at startup → **Disable**.
3. Usually safe to disable: launchers you don't use daily, chat apps, cloud sync, music apps, updaters, trial software.
4. **Keep enabled:** audio (Realtek, Dolby), touchpad, security/antivirus, NVIDIA, and Lenovo Vantage / Legion Space. Disabling doesn't uninstall anything; you can still open the app any time.

5.2 Close these before you play

Type of app	Examples	Why close it
VPN	Any VPN client	Adds network delay (ping) and uses CPU. Close it unless you need it.
Cloud sync	Google Drive, OneDrive, Dropbox	Can start uploading or scanning files mid-game. Pause or quit.
Extra game launchers	Launchers for games you aren't playing	Each runs background updaters and overlays.
RGB / vendor utilities	Lighting apps, extra hardware tools	Some check hardware constantly. Keep only one control app (Vantage / Legion Space).
Web browser	Chrome, Edge, Firefox	Lots of tabs use RAM and GPU, especially video tabs.

5.3 Check the Task Manager GPU column

1. In Task Manager → **Processes**, right-click a column header and make sure **GPU** is ticked.
2. With no game running, click **GPU** to sort. Anything using more than a few percent at idle deserves a look.
3. If it's an NVIDIA process, follow Chapter 2.6. If it's another app, close it or disable it from startup.

SKIP "DEBLOAT" SCRIPTS AND REGISTRY TWEAKS

Tools that delete parts of Windows or edit the registry can break updates and security, for little or no real gain. This guide only uses settings you can easily undo.

6 RAM and Storage

Give your games room to breathe

Difficulty: Easy | Time: 10–20 min

6.1 Keep 15–20% of your SSD free

An SSD *(solid-state drive: your laptop's fast storage)* slows down when it's nearly full, and games need free space for shader caches and updates. On a 1 TB drive, aim to keep roughly 150–200 GB free. To check: **File Explorer** → **This PC** → look at the bar under **C:**.

6.2 Free up space safely

1. **Storage Sense:** Settings → System → **Storage** → turn on **Storage Sense**. It cleans temporary files automatically.
2. **Disk Cleanup:** Start → type **Disk Cleanup** → choose C: → **Clean up system files** → tick Temporary files, Delivery Optimization files, and Windows Update Cleanup → **OK**.
3. **Leftover game folders:** uninstalling a game often leaves mods, saves, and caches behind. Check `Documents\My Games` and your game library folders (such as `Steam\steamapps\common`) for games you've removed. **Back up any saves you want** before deleting.
4. **Uninstall unused apps:** Settings → Apps → Installed apps → sort by **Size**.

6.3 Installing big modlists (Nolvus and similar)

- Use a **short install path** at the top of a drive, like `C:\Nolvus`. Avoid Program Files, the Desktop, Downloads, and any OneDrive folder.
- If space is tight, **turn off "archiving"** in the modlist installer. Archiving keeps a copy of every downloaded mod file for reinstalls. It's handy, but it can take a very large amount of extra space.
- Check the modlist's official space requirement **before** you start, and leave a safety margin.

6.4 RAM: check dual channel and free memory

RAM *(short-term working memory for apps and games)* runs fastest in dual channel *(two memory sticks working side by side, doubling bandwidth)*.

1. Task Manager → **Performance** → **Memory**. Look at **Slots used**: "2 of 2" means dual channel. Many 32 GB laptops ship as 2 × 16 GB, which is ideal.
2. If it says "1 of 2", the laptop runs in single channel. Adding a matching second stick (same size and speed) can help. That's a hardware upgrade, so check your manual or ask a technician.
3. Before playing, **close browser tabs** and other heavy apps. Note the **Available** memory for the Chapter 8 checklist.

7 Cooling and Airflow

A cooler laptop holds its speed longer

Difficulty: Easy | Time: 10 min

IN PLAIN ENGLISH

When chips get too hot, the laptop slows them down to protect itself. This is thermal throttling *(automatic slowdown due to heat)*, and it shows up as FPS that starts high and drops after a few minutes.

Do

- Use a **hard, flat surface** like a desk or table.
- Try a **cooling pad** or a stand that raises the back.
- Keep the **rear and side vents clear**; that's where hot air leaves.
- Clean vents every few months with **short bursts of compressed air** (laptop off and unplugged).
- **Monitor temperatures** while gaming (Chapter 8).

Don't

- Game on a bed, couch, blanket, or your lap. Fabric blocks the intake vents.
- Push the back of the laptop against a wall.
- Let the fans spin freely from the air blast while cleaning.
- **Repaste** (replace the thermal paste) unless you're experienced. Many laptops use liquid metal or delicate pads, and opening the case can affect your warranty.

7.1 What temperatures are normal?

Gaming laptops run hot by design. Use this as a rough guide, and check your laptop maker's published limits for your model.

What you see while gaming	What it means
GPU in the 70s to low 80s °C	Typical for a gaming laptop under load.
CPU in the 80s to 90s °C	Common for modern Intel laptop chips, which are designed to boost up to their limit.
FPS drops after 5–10 minutes, temps very high	Possible throttling. Check the surface, vents, dust, and that Performance mode is on.

TIP: A FRAME CAP ALSO COOLS

Capping FPS (Chapter 2.3) stops the GPU from working harder than your screen can show. Less heat and quieter fans, for free.

8 Before / After Performance Checklist

Measure, tune, measure again

Difficulty: Easy–Medium | Time: about 5 min per game, twice

8.1 How to measure

Option A: NVIDIA overlay (easiest)

1. Temporarily turn on the NVIDIA Overlay (NVIDIA app → Settings → Features).
2. In a game, press `Alt + R` to show the performance stats, or `Alt + Z` → **Statistics** to choose what's shown.
3. Read FPS, GPU temperature, and GPU usage.
4. Turn the overlay off again afterward if you don't use it.

Option B: MSI Afterburner (more detail)

1. Install **MSI Afterburner**; it includes **RivaTuner Statistics Server** for the on-screen display.
2. Settings → **Monitoring**: tick Framerate, Frametime, GPU and CPU temperature, and "Show in On-Screen Display".
3. Use its **benchmark** hotkeys to record average FPS and **1% lows**.
4. Don't touch its overclocking sliders.

Average FPS *(the typical frame rate)* tells you how fast a game runs. 1% lows *(the frame rate during the slowest 1% of moments)* tell you how **smooth** it feels, because they reveal stutters.

MAKE IT A FAIR TEST

- Use the same spot in the game, the same settings, and the same 1–2 minutes of play both times.
- Cyberpunk 2077 has a built-in **Benchmark** button in its graphics menu. Use it.
- Play a few minutes first so shaders finish compiling (Chapter 10).

8.2 System checks (idle, no game open)

Measurement	Before	After	Where to find it
GPU usage at idle (%)			Task Manager → Performance → GPU
Free (Available) RAM (GB)			Task Manager → Performance → Memory
Free disk space on C: (GB)			File Explorer → This PC
Startup apps enabled (count)			Task Manager → Startup apps
NVIDIA driver version			NVIDIA app → Drivers

8.3 In-game results

Game	Avg FPS before	Avg FPS after	1% low before	1% low after	GPU °C before / after	CPU °C before / after
Skyrim SE / Nolvus						
Fallout 4						
Cyberpunk 2077						
Other: _____						
Other: _____						

Notes (settings, resolution, stutters noticed): _____

9 · 15-Minute Quick-Start Card

Do the Chapter 1 updates first. Plug in the charger. Then tick each box.

15 MIN

1 ☐ **Plug in & Performance mode**

Connect the charger. Press **Fn + Q** until the power light is **red**. Other brands: pick Performance/Turbo in the brand's app.

Ch. 4.4 · 1 min

2 ☐ **GPU mode: dGPU / Discrete**

In Vantage / Legion Space, set GPU mode to **dGPU only** for gaming (Hybrid for battery). **Restart.**

Ch. 4.4 · 2 min

3 ☐ **NVIDIA 3D settings**

Manage 3D settings: **Prefer maximum performance** · Shader cache **10 GB** · Texture filtering **High performance** · Low latency **On** · Refresh **Highest** · Threaded **Auto**.

Ch. 2.1 · 3 min

4 ☐ **G-Sync recipe**

G-Sync **On** · V-Sync **On** in Control Panel · V-Sync **Off** in games · cap **~3 fps below** refresh (162 at 165 Hz), or use Reflex.

Ch. 2.3 · 2 min

5 ☐ **DLSS overrides**

NVIDIA app → Graphics: model **Latest** · Super Resolution **Preset K** · Frame Gen **2x** (only if base fps is ~60+) · Ray Reconstruction **On**.

Ch. 2.4 · 1 min

6 ☐ **Kill hidden GPU load**

Turn off Automatic Tuning, RTX Video SR/HDR, and the Overlay + Instant Replay if unused. Task Manager GPU column at idle should be near 0%.

Ch. 2.5–2.6 · 2 min

7 ☐ **Windows settings**

Power mode **Best performance** (or Ultimate / High Performance) · **Game Mode** On · **Hardware-accelerated GPU scheduling** On.

Ch. 4.1–4.3 · 2 min

8 ☐ **Trim background apps**

Task Manager → Startup apps: disable extras. Before playing, close VPN, Drive/OneDrive, extra launchers, RGB tools, and browser tabs.

Ch. 5 · 1 min

9 ☐ **Game profiles & space**

Nolvus: **2K** textures · Fallout 4: **iPresentInterval=1**, cap ≤ 141 · Cyberpunk: DLSS **Quality** + Reflex. Keep **15–20%** of C: free.

Ch. 3 & 6 · 1 min

10 ☐ **Airflow & test**

Hard flat surface, rear vents clear. Play 10 minutes (shaders compile), then fill in the **After** column of the Chapter 8 checklist.

Ch. 7–8 · ongoing

NEVER

interrupt a BIOS update · uncap Fallout 4 · enable overclocking as a beginner · run "debloat" scripts · block the rear vents.

Tip: print this page and keep it by your laptop. Repeat steps 6 and 8 after every driver update.

10 What to Expect & Disclaimer

Realistic results, honest trade-offs

No specific FPS number is promised: every laptop, game version, driver, and mod setup differs. Here's what a correct tune-up **typically** delivers, and what it costs.

10.1 What usually improves

Improvement	Why it happens
Smoother frame pacing	Frame pacing *(how evenly frames are spaced)* improves with G-Sync, V-Sync, and a proper frame cap, removing judder and tearing even when the FPS number barely changes. Good airflow and a cap also keep long sessions steadier.
Fewer stutters once the shader cache builds	A larger shader cache means games don't re-prepare shaders each session. The improvement shows from the second session onward.
Often a noticeable FPS gain	Mostly from **removing background load** (like the idle GPU usage in 2.6) and using the **correct power and GPU modes**. How much depends on how un-tuned the laptop was to start with.

10.2 The trade-offs

LOUDER FANS

Performance mode spins the fans faster. That's normal and means it's working. Use Balanced or Quiet for light games and late nights.

SHORTER BATTERY LIFE

dGPU-only mode and maximum-performance settings drain the battery faster. Switch to Hybrid + Battery saver when traveling (restart needed).

THE FIRST LAUNCH MAY STUTTER

After a driver update or new game install, the first few minutes can stutter while shaders compile. Some games show a "Compiling shaders" screen. Let it finish; the next launch is smoother.

10.3 Keeping it tuned

Update the NVIDIA driver about monthly (then re-check 2.5, since updates can switch features back on), check Vantage / Legion Space for BIOS updates every few months, and clean the vents every 3–6 months.

Disclaimer

This guide is an independent product and is **not affiliated with, endorsed by, or sponsored by NVIDIA, Lenovo, Intel, or Microsoft**, or by any game developer or modlist author. All product names and trademarks belong to their owners. Menu names and options can change with software updates. **Results vary** with hardware, software versions, game updates, and mods, and no specific performance gain is guaranteed. You apply these changes **at your own risk**. Always **back up your files before a BIOS update** or other major change, and create a Windows restore point before you begin. If you're unsure about a step, stop and contact your laptop maker's support.

A Glossary

Every technical term in this guide, in plain English

1% lows	The frame rate during the slowest 1% of gameplay. Low numbers mean stutters.
BIOS	Built-in firmware that starts the laptop before Windows and controls fans and power limits.
Clean install	Installing a driver after wiping the old one and its settings.
dGPU / iGPU	Discrete GPU (the separate NVIDIA card) vs. integrated GPU (small graphics built into the CPU).
Driver	Software that lets Windows talk to a piece of hardware.
ENB	A graphics add-on for Skyrim and Fallout that changes lighting and effects; includes a frame limiter.
Frame cap / limiter	A maximum FPS setting. Reduces heat and keeps G-Sync active.
Frame pacing	How evenly frames are spaced. Even pacing feels smooth.
G-Sync / VRR	Variable refresh rate: the screen refreshes when a new frame is ready, removing tearing and judder.
HAGS	Hardware-accelerated GPU scheduling: the GPU manages its own work queue. Needed for Frame Generation.
Latency	The delay between your input and seeing the result on screen.
Modlist	A curated, pre-tested bundle of mods that installs together, such as Nolvus.
Overclocking	Running a chip faster than its stock speed. Not recommended for beginners.
RAM	Short-term working memory used by apps and games.
Ray Reconstruction	DLSS feature that cleans up noisy ray-traced lighting.
1080p / 1440p	Screen resolutions of 1920×1080 and 2560×1440 pixels. Higher is sharper but harder to run.
Chipset driver	Software that helps Windows manage the motherboard, power, and ports.
CPU	The processor, the laptop's main "brain".
DLSS	NVIDIA's AI upscaling: renders at a lower resolution and rebuilds a sharp image for more FPS.
Dual channel	Two RAM sticks working together for double the memory bandwidth.
FPS	Frames per second: how many images the game draws each second.
Frame Generation	DLSS feature where AI inserts extra frames between real ones. Best when base FPS is about 60+.
Game Ready driver	NVIDIA's gaming-focused driver, tuned for new releases.
GPU	The graphics card, the chip that draws the game.
Hz (refresh rate)	How many times per second the screen redraws, e.g. 165 Hz.
LOD bias	Controls how sharp distant textures look.
MUX switch	Hardware that chooses which GPU drives the screen. Changing it needs a restart.
Overlay	On-screen info or menus drawn over a game (NVIDIA, Steam, Discord).
Ray tracing (RT)	Realistic simulated lighting, shadows, and reflections. Demanding.
Reflex	NVIDIA's lag-reducing feature, built into many games.

Restore point	A saved snapshot of Windows settings you can roll back to.
SSD	Solid-state drive, the laptop's fast storage.
Super Resolution	The upscaling part of DLSS. Presets (like K) are different AI models.
Thermal throttling	Automatic slowdown when chips get too hot.
VRAM	The graphics card's own memory (8 GB on an RTX 5070 Laptop GPU).
Shader / shader cache	Small programs that draw surfaces and lighting; the cache saves them so they aren't rebuilt each time.
Stutter	A brief hitch or freeze during gameplay.
Tearing	A horizontal split in the image when frames and screen refreshes are out of sync.
V-Sync	Syncs frames to the screen's refresh to prevent tearing.
Vortex	A popular mod manager app.